



THE UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
MBINGA DISTRICT COUNCIL
FORM IV DISTRICT PRE MOCK EXAMINATIONS MARCH 2026



041

BASIC MATHEMATICS

Time: 3:00 HOURS

Tuesday 17th March 2026 A.m.

INSTRUCTIONS

1. This paper consists of sections A and B with a total of **fourteen (14)** questions.
2. Answer **all** questions. **wazaelimu.com**
3. Each question in section A carries **six (06)** marks while each question in section B carries **ten (10)** marks.
4. All necessary working and answers for each question must be shown clearly.
5. Mathematical tables and non-programmable calculators may be used.
6. All communication devices and any unauthorized materials are not allowed in the examination room.
7. Write your **examination number** on every page of your answer sheets.

SECTION A (60 Marks)

Answer **all** questions in this section

1.(a) Frola receives a salary of 960 000 Tanzanian shillings every month. She spends 40 % for buying food, 20 % for paying school fees for her children, 15 % for buying other home expenditures and the remaining percentage for saving. Find the amount of money that she saves.

b) Four sugar canes with lengths of 70 cm, 84 cm, 105 cm and 119 cm are cut into pieces of the same length. Find the greatest possible length for these pieces if no sugar cane is left over.

2.(a) Solve for x if $(\frac{1}{81})^{-6x} \times 81 = \sqrt{9}$

(b) Find the value of x given that $\log (3x + 2) - 1 = \log (x - 4)$

3. (a) In a village of 40 people, all the people speak Kiswahili or English or both. The number of people who speak both Kiswahili and English is three times the number of people who speak English only. Furthermore, the number of people who speak Kiswahili only is 6 more than twice the number of people who speak English only. If 4 people speak neither Kiswahili nor English, find the number of people who speak (i) Kiswahili only.

(ii) Both Kiswahili and English.

(c) A family has 2 children. Using a tree diagram, find the probability that one child is a boy and the other is a girl.

4. (a) Find the value of y so that the straight line joining the points $(x, -5)$ and $(2, 1)$ is parallel to the straight line joining the points $(1, 5)$ and $(-2, -1)$.

(d) If $\underline{a} = (4, 3)$, $\underline{b} = (-4, 1)$ and $\underline{c} = (2, 5)$, determine which of the vectors $\underline{a} + 2\underline{b}$ and $3\underline{a} + \underline{c}$ is longer than the other.

5.a) Find the area of triangle ABC given that $AB=8$ cm, $BC=11$ cm and angle $ABC = 30^\circ$

b) The sides of a triangle are 4 cm, 5 cm, 6 cm. If the longest side of a similar triangle is 18 cm, find the height of other sides

6 a) A lorry weighs 7.5 tones. If the total mass of the bricks carried by the lorry is 5 tones 400 kilograms 50 grams, what is the total weight of the lorry and bricks? Give your answer in kilograms.

b) Three tailors can sew 15 clothes in 5 days. How long will it take 5 tailors working at the same speed to sew 20 clothes?

7. (a) A shopkeeper bought 10 bags of rice for 200 000 Tanzanian shillings. If 4 bags were sold at 30 000 Tanzanian shillings each, and the remaining bags were sold at 25, 000 Tanzanian shillings each, what was the percentage profit?

b). On 1st July 2022, Neema started a business with the capital in cash of Tsh 1 000 000 and made the following transactions:

July 2	Purchased goods for cash	Tsh 700 000
July 3	Sold goods for cash	Tsh 500 000
July 6	Purchased goods for cash	Tsh 300 000
July 10	Bought goods for cash	Tsh 235 000
July 15	Paid cash for rent	Tsh 110 000
July 26	Paid cash for wages	Tsh 55 000
July 28	Sold goods for cash	Tsh 310 000

By using these transactions, prepare the cash account.

8.(a) The starting salary of a worker in a certain company is Tsh 2 400 000 per annum. If the annual increment is Tsh 20 000, what is the total amount of money earned by the worker in four years?

b).Paula invested a certain amount of money in a saving account which provides an interest rate of 10 % compounded annually. After two years, she had a total of Tsh 50 000.

How much did she invest at the beginning?

How much did she receive as interest at the end of the second year?

9. (a) A rectangular field has a diagonal of 100 metres. If its length is 80 meters, find the width of the field.

b).From the top of a building which is 50 m high, the angle of elevation of a car parked on the ground is 30° . How far is the car from the base of the building? Leave your answer in surd form.

10 .a)Julius is four times older than Amina. Five years ago, the sum of their ages was 50 years.

How old are they now?

b)A trapezium has an area of $(2x^2 - 8x + 6)$ square units. If the parallel sides are $(2x + 3)$ units and $(2x - 7)$ units long, find its height.

SECTION A (40 Marks)

Answer **all** questions in this section

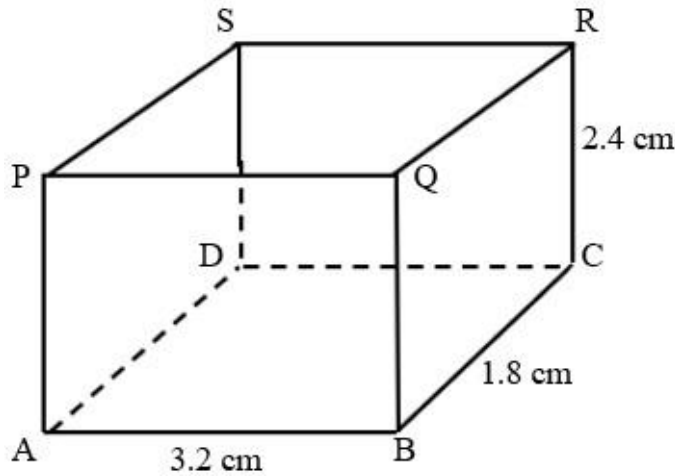
11.The ages of children who suffered from Cholera in Kagabilo village were given in the table below.

Age (years)	0 – 4	5 – 9	10 – 14	15 – 19	20 – 24
Number of children	12	15	12	8	3

a)Find the mean number of children who suffered from Cholera by using assumed mean method.

b)Find the mode and median.

12. (a) The figure below is a rectangular box ABCDPQRS where $\overline{AB} = 3.2 \text{ cm}$, $BC = 1.8 \text{ cm}$ and $\overline{CR} = 2.4 \text{ cm}$. Calculate the angle between the diagonal $\overline{AR}$ and the base ABCD.



- b) A ship sails from A ($0^\circ, 20^\circ \text{ W}$) to B ($10^\circ \text{ N}, 20^\circ \text{ W}$) at 16 knots. If it leaves point A at 8 a.m. on Tuesday, at what time will it arrive at point B?

13. Anna is running a grocery at a certain school. She is selling pieces of cake and soda. Each piece of cake costs 1 500 shillings and each bottle of soda costs 500 shillings. In one afternoon, Anna made a total of 255 500 shillings by selling a total of 225 pieces of cake and soda. Use the matrix method to find the number of pieces of cake and bottles of soda sold.

- b) Find the image of point A (4, 2) after a reflection in the line $y + x = 0$ followed by an anticlockwise rotation through 90° about the origin.

14. (a) The function f is given as $f(x) = 3x^2 - 1$. Calculate the domain and range of the inverse of f .

$$f^{-1}\left(\frac{1}{3}\right)$$

- b) A farmer has 10 acres of land on which he can grow either mangoes or oranges. He has 59 working days available during cultivation season. Mangoes require 5 days per acre of labour while oranges require 8 days of labour per acre. The net profit from mangoes per acre is 30 000 shillings and that of oranges is 45 000 shillings. How many acres of each crop should he plant to maximize his profit

